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Rem:-
* Practice more series problem.
* Remember your mistakes.

30.8.25

TEST

SECTION-A

1] a → 9name ✗ (b) identifier name never starts with a digit.
2] d → false ✓
3] a → 3 ✓
4] b → char ✗ String Characters stored in String character stored in char
5] a → Math.sqrt() ✓
6] c → 15 ✓
7] c → ++i increases i before use ✓
8] c → -128 to +127 ✓
9] b → switch ✓
10] b → 8 ✓

(8)

SECTION-B

9.5/10

11] int x = 87

x++ + ++x ;

~~5+6+6~~

= 5 + 7 ;

= 12 ✓

x = 7 ✓

OUTPUT

12

7

Write like this

Output:
12
7

4.5

12] double a = Math.sqrt(81) + Math.pow(2,5) - Math.abs(-10)

= 9.0 + 32.0 - 10

= 31.0

OUTPUT

31.0 ✓

(5)

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SECTION-C

13]

IMPLICIT	EXPLICIT
• Requires no user interfere	• Requires user inteference
○ No data loss. ✓	○ Data loss is possible. ✓
• int a, b;	int c;
double c = a+b;	~~Math.sqrt(6221)~~
// this will result in double ✓	c=(int) Math.sqrt(6221)
	// now the user has explicitly type casted, so it will result in int ✓

(2)

Conversion from lower data type to higher data type only.

Conversion from higher data type to lower data type.

14] y = 3 4 21 x = 5 6 7

```
y+= x++ + y++ + x++ + y ;
=> y = y+(x++ + y++ + x++ + y) ;
   = 3+( 5 + 3 + 6 + 4) ;
   = 3+18 ;
=> y = 21
```

✓ (4)

15]

MATH.FLOOR()	MATH.CEIL()
• Gives the lowest int value ✓	Gives the highest int value ✓
• Math.floor (16.001)	Math.ceil (16.523)
// Output would be 16 ✓	// Output would be 17 ✓

(4)

16] a = 10 9 16 15 b = 8 8 8 11

a = ++b + a--;

⇒ a = 6 + 10;

= 16

b = a-- - --b;

⇒ b = 16 - 5

= 11

∴ a = 15

∴ b = 11

4

17] $\dfrac{(x^2+y^2)z + |x-y|}{(x^2+y^2)\sqrt{2}} + |x-y|$

4

Ans -

[(x * x) + (y * y)] * z + (x-y) / [((x * x) + (y * y)) * (*) Math. sqrt (z)] + (x-y)

P.T.O

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SECTION-D

18)

```
import java.util.Scanner;
public class Vote
{
  public static void main (String args[])
  {
   Scanner sc = new Scanner (System.in);
   System.out.println("If Enter age of person:");
   int age = sc.nextInt();
   if (age <= 18)
    {
     System.out.print (" Not Eligible to Vote");
    }
   else if (age >= 18)
        {
         System.out.print (" Eligible to Vote ");
        }
   else
      {
       System.out.print (" ERROR ");
      }
  }
}
```

9 ✓

19]
```
import java.util.*;
public class Grades
{
 public static void main (String args[])
 {
  Scanner sc = new Scanner (System.in);
  System.out.print ("\n Enter marks : ");
  int mks = sc.nextInt();
  if (mks >= 90 && mks <= 100)
  {
   System.out.print ("\n Grade A");
  }
  else if (mks >= 75 && mks <= 89)
     {
      System.out.print ("\n Grade B");
     }
  else if (mks >= 50 && mks <= 74)
     {
      System.out.print ("\n Grade C");
     }
  else if (mks < 50)
      {
       System.out.print ("\n Grade D");
      }
   else
      {
       System.out.print ("\n ERROR");
      }
  }
}
```

10

20]
```
import java.util.*;
public class Square_cube
{
  public static void main (String args[])
  {
    Scanner sc = new Scanner (System.in);
    System.out.print("\n If Press 1 to calculate square of a no.
                          and 2 to calculate cube of a no.");
    System.out.print("\n Enter your choice 1-2-3. 3 is to
                          exit : ");
    int ch = sc.nextInt();
     switch (ch)
     {
      case 1 : {
          System.out.print("\n Enter a no.: ");
          int x = sc.nextInt();
          int a = Math.pow(x, 2);
          System.out.print("\n Square of "+x+" is = "+a);
      break;
          }
      case 2 : {
          System.out.print("\n Enter a no.: ");
          int y = sc.nextInt();
          int b = Math.pow(y, 3);
          System.out.print("\n Cube of "+y+" is = "+b);
      break;
          }
      case 3 : {
          System.out.print("\n Sir pls give marks ! Bye !");
      break;
          }
      default : {
          System.out.print("\n ERROR ");
      break;
    }
  }
}
```

10

21]
```
import java.util.*;
public class Series
{
  public static void main (String args[])
  {
    Scanner sc = new Scanner (System.in);
    System.out.print ("\n Enter ending term : ");
    int end = sc.nextInt ();
    int sum=0;
    for( int i = 1 ; i <= end ; i+=1)
    {
      sum += i ;
      System.out.print (i + " + ");
    }
      sum+= end ;
      System.out.print (end+" = " + sum);
  }
}
```

22]
```
import java.util.*;
public class Prime_check
{
  public static void main (String args[])
  {
    Scanner sc = new Scanner (System.in);
    System.out.print ("\n Enter a no.");
    int n = sc.nextInt ();
    int cnt =0;
    for( int i = 1 ; i <= n ; i++)
    {
      if ( n % i == 0)
      {
        cnt++;
      }
    }
    if (cnt == 2)
      System.out.print ("\n PRIME");
    else if (cnt > 2)
      System.out.print ("\n COMPOSITE");
    else
      System.out.print ("\n ERROR ");
  }
}
```